

1. O/N/22/21/Q5

Fig. 5.1 is a diagram that shows the number of particles in a charged atom (ion) of beryllium (Be).

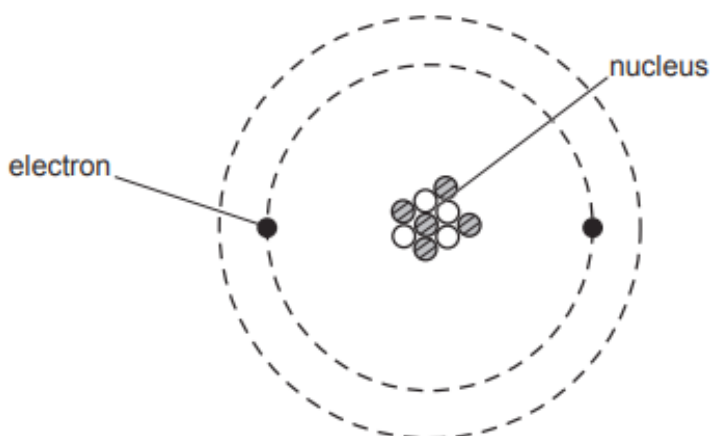


Fig. 5.1

- (a) The charge on an electron is $-1.6 \times 10^{-19} \text{ C}$. The charge on the ion is $+3.2 \times 10^{-19} \text{ C}$.

(i) Deduce, from the charge on the ion, the number of protons in its nucleus.

Show your working.

number of protons = [2]

(ii) Write down, in nuclide notation, the symbol for the nucleus of this charged atom.

..... [1]

- (b) The isotope in Fig. 5.1 is the only stable isotope of beryllium.

Explain the term 'isotope'.

.....

 [2]

- (c) Explain what prevents electrons in an atom from escaping.

.....

 [2]

[Total: 7]

2. M/J/15/22/Q8

Two isotopes of hydrogen are written as ${}^1_1\text{H}$ and ${}^2_1\text{H}$.

- (a) Complete Fig. 8.1 to show the number of protons and neutrons in one nucleus of each of these isotopes of hydrogen.

	number of protons	number of neutrons
${}^1_1\text{H}$		
${}^2_1\text{H}$		

Fig. 8.1

[2]

- (b) Explain, using ideas about electrons, neutrons and protons, why atoms of ${}^1_1\text{H}$ and ${}^2_1\text{H}$ are uncharged.

.....

.....

.....

.....

.....

.....

..... [3]

3. O/N/12/22/Q8

The nuclei of four neutral atoms are represented using nuclide notation as:



- (a) Neutral atoms also contain electrons. State which of the four nuclei is surrounded by the greatest number of electrons.

..... [1]

- (b) From these four nuclei, state the **two** nuclei that have the same number of

- (i) protons,

..... [1]

- (ii) nucleons,

..... [1]

- (iii) neutrons.

..... [1]

4. O/N/09/Q8

(a) State the number of

(i) electrons in a neutral atom of the isotope ${}^7_3\text{Li}$,

number of electrons = [1]

(ii) nucleons in an atom of the isotope ${}^{208}_{82}\text{Pb}$,

number of nucleons = [1]

(iii) protons in an atom of the isotope ${}^{23}_{11}\text{Na}$.

number of protons = [1]

(b) An atom of the isotope chlorine-35 contains 17 protons.

For an atom of the isotope chlorine-37, state the number of

(i) protons,

number of protons = [1]

(ii) neutrons.

number of neutrons = [1]